SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 1.1: Real Numbers — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.1: Real Numbers
Driving Question	DRIVING QUESTION: Why can some numbers be written as exact fractions and others cannot — and how does knowing a number's "type" help us solve everyday problems in Kenya?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **What Kind of Number Is It? — Launching the Pizza Phenomenon**	Students examined a real-world anchoring phenomenon involving equal sharing of chapati/pizza portions among different group sizes (e.g., sharing 12 pieces among 2, 3, 4, or 5 people). They recorded which divisions came out 'perfectly whole' and which did not, noticing that some results were neat whole numbers while others produced leftovers or fractions. Teachers should probe prior knowledge by asking students to list numbers they use daily (prices at a Nairobi market, running distances, maize yields in the Rift Valley) and sort them informally.	Students discovered that whole numbers from 1 to 12 behave differently depending on how they can be divided. Numbers divisible by 2 are even; those that are not are odd. Some numbers (primes) can only be divided evenly by 1 and themselves, while others (composites) have additional factors. This classification is the first step toward understanding why some numbers produce clean fractions and others do not — directly connecting to the driving question.	The teacher should use this lesson to establish the DQB anchor: post the driving question prominently and invite students to record what they already believe and what they wonder. Pedagogical note: Emphasise that 'type' is not just a labelling exercise — knowing a number is prime, for example, means it cannot be broken into smaller equal groups, which has direct implications for fair sharing in Kenyan market contexts (e.g., splitting a 7 kg bag of ugali flour equally). Cold-call at least 3 students to explain their sorting rationale before formalising definitions.
Lesson 2 **Sorting the Number World: Odd, Even, Prime, and Composite — and the Pizza That Started It All**	Students systematically sorted whole numbers using factor trees and divisibility rules, recording results in a class table. They observed patterns — for example, all even numbers greater than 2 are composite, and 2 is the only even prime. Teachers should have students use physical counters (e.g., dried maize kernels or beans, which are readily available in Kenyan classrooms) to physically arrange numbers into equal rows, making the	Whole numbers are classified as odd (not divisible by 2), even (divisible by 2), prime (exactly two factors: 1 and itself), or composite (more than two factors). Students learned that these categories are mutually exclusive in meaningful ways (e.g., 1 is neither prime nor composite — a common misconception to address explicitly). This builds the foundation for understanding rational numbers, since composite numbers are more easily	Teachers should return to the anchoring phenomenon: 'When you divided 12 chapati pieces among 5 people, what happened — and what type of number is 5?' This links prime numbers directly to the driving question about exact fractions. Pedagogical note: Address the misconception that 'bigger numbers are always composite' — introduce a prime like 97 or 101. Allow 5 minutes at lesson end for students to update their DQB with new questions or

	concept of divisibility concrete and tactile.	expressed as products of fractions.	revised answers. Ensure model diagrams now include a basic number-type sorting chart.
Lesson 3 **Reciprocals of Real Numbers — Division, Tables, and Calculators**	Students used calculators and division tables to compute 1 ÷ n for various values of n (whole numbers, fractions, and decimals). They recorded the decimal outputs and observed that some reciprocals terminate (e.g., 1 ÷ 4 = 0.25) while others repeat endlessly (e.g., 1 ÷ 3 = 0.333…) and others appear to be non-repeating (e.g., 1 ÷ √2). Teachers should direct students to use Kenyan currency examples: 'If 1 litre of milk costs Ksh 1, how much is one-fifth of a litre worth?'	The reciprocal of any non-zero real number n is 1/n — it answers the question 'how many of this piece fit into 1 whole?' The reciprocal of a rational number is always rational; the reciprocal of an irrational number is also irrational. Zero has no reciprocal because division by zero is undefined. Students should leave this lesson able to compute and interpret reciprocals in context (e.g., a Kericho tea farmer dividing a plot into n equal sections).	Pedagogical note: The reciprocal concept is often confused with 'the negative' of a number. Teachers should explicitly contrast: the reciprocal of 3 is 1/3 (not −3). Use wait time of at least 10 seconds after posing the question 'What happens when you multiply a number by its reciprocal?' before cold-calling. Guide students to discover that n × (1/n) = 1, establishing the multiplicative inverse identity. Connect back to the driving question: reciprocals of rational numbers are rational — does that hold for irrational numbers?
Lesson 4 **Rational or Irrational? Placing Real Numbers on the Number Line**	Students plotted a set of numbers — including integers, common fractions, terminating decimals, repeating decimals, √2, √3, and π — on a large number line displayed on the classroom wall or drawn on the ground outdoors. They observed that rational numbers could be placed at exact, nameable positions, while irrational numbers (e.g., π ≈ 3.14159…) could only be approximated. Teachers should use the context of measuring distances along a Nairobi running track or the length of a shamba in Kisumu to make placement meaningful.	Real numbers are classified as rational (expressible as p/q where p and q are integers and q ≠ 0; their decimals terminate or repeat) or irrational (cannot be written as p/q; their decimals are non-terminating and non-repeating). Together, rational and irrational numbers form the complete set of real numbers. Students learned that √4 = 2 is rational, but √2 is irrational — a critical distinction that directly answers the driving question.	Pedagogical note: This is the conceptual core of the unit — teachers should spend extra time here. Use a Venn diagram on the board to map number types (natural → whole → integers → rationals → reals, with irrationals sitting outside the rational set but inside reals). Cold-call at least 4 students to classify a number and justify their reasoning. Connect to the DQB: 'Has our driving question been answered yet? What evidence do we now have?' Students should update their models to include the rational/irrational distinction on their number classification diagram.
Lesson 5 **Flipping Numbers: Introducing Reciprocals by Division**	Students engaged in a hands-on activity using folded paper strips or cut pieces of mkate (bread) to physically demonstrate reciprocals. For example, if one strip represents 1 whole, folding it into 3 equal parts shows that each part is 1/3 — the reciprocal relationship to 3. They recorded: 'If I have 1/3 of a piece, how many do I need to make 1 whole? → 3.' This physical manipulation reinforces the meaning of reciprocal as 'how many fit into 1.'	The reciprocal of a number n is 1 ÷ n (equivalently, b/a for a fraction a/b), and it represents how many of that quantity fit into 1 whole. The reciprocal exists for all real numbers except zero. For fractions, students flip the numerator and denominator (e.g., the reciprocal of 3/4 is 4/3). This lesson deepens understanding from Lesson 3 by grounding reciprocals in physical, proportional reasoning rather than purely procedural calculation.	Pedagogical note: Although this lesson revisits reciprocals (introduced in Lesson 3), the focus here shifts from computation to conceptual meaning through physical modelling — do not treat this as repetition but as deepening. Teachers should explicitly ask: 'Is the reciprocal of an irrational number also irrational? How does that connect to our driving question?' Allow students to debate this in pairs for 3 minutes before whole-class discussion. Update the DQB to reflect any new insights about how reciprocals interact with rational vs. irrational numbers. Ensure model diagrams now show the reciprocal

			relationship with arrows linking a number and its inverse.
Lesson 6 **Lesson 6**	Students investigated operations (addition, subtraction, multiplication, division) applied to combinations of rational and irrational numbers, recording whether results remained rational or became irrational. For example: rational + rational = rational; rational + irrational = irrational; irrational × irrational can be rational (e.g., $\sqrt{2} \times \sqrt{2} = 2$). Teachers should use Kenyan measurement contexts — mixing volumes of uji (porridge) or combining lengths of fencing wire on a Rift Valley farm — to frame each operation meaningfully.	The set of real numbers is closed under addition and multiplication, but mixing rational and irrational numbers in operations does not always preserve the type in predictable ways. Students consolidated their understanding of the number hierarchy: natural numbers ⊂ whole numbers ⊂ integers ⊂ rational numbers ⊂ real numbers, with irrational numbers filling the 'gaps' on the number line between rationals. This lesson answers the driving question more fully by showing that knowing a number's type predicts the outcome of operations.	Pedagogical note: This lesson is critical for consolidation — it surfaces and corrects the common misconception that 'irrational × irrational is always irrational.' Teachers must provide a counterexample ($\sqrt{2} \times \sqrt{2} = 2$) explicitly. Use wait time of 15 seconds after posing this counterexample before asking students to update their mental models. Cold-call at least 3 students to revise their earlier DQB predictions. Students should significantly update their cumulative models in this lesson — the diagram should now show subsets, operations, and the rational/irrational boundary with annotated examples drawn from Kenyan daily life.
Lesson 7 **Final Explanation: Putting It All Together — Real Numbers, Reciprocals, and Everyday Life in Kenya**	Students reviewed their completed Summary Tables, DQB entries, and model diagrams from all previous lessons, identifying the progression of their understanding. They then applied their full understanding to a culminating real-world task: analysing a scenario such as a Mombasa fish vendor dividing a catch, a Kericho tea estate calculating plot areas involving $\sqrt{3}$ km boundaries, or a Nairobi matatu operator splitting fares — determining what types of numbers arise and how to work with them accurately.	Real numbers are classified as rational (expressible as p/q, where $q \neq 0$) or irrational (non-terminating, non-repeating decimals such as $\sqrt{2}$ and π). Every non-zero real number has a reciprocal (its multiplicative inverse). Knowing whether a number is rational or irrational determines how precisely it can be expressed and predicts behaviour under operations — directly and fully answering the driving question. Students can now articulate why fractions work for some measurements (fair market shares, equal land divisions) but not others (circular distances involving π, diagonal lengths involving $\sqrt{2}$).	Pedagogical note: This lesson is the payoff — teachers should facilitate a full-class DQB review, reading back the original driving question and asking students to answer it collaboratively using evidence gathered across all 7 lessons. Cold-call at least 5 students to contribute one piece of evidence each. Students should present their final revised models (individually or in pairs), showing the complete real number classification system with Kenyan examples annotated at each level. Conclude by asking: 'What questions about numbers do you still have?' — record these on the DQB as a bridge to future strands (e.g., surds, logarithms, complex numbers in later grades).